

Guidelines for Human Evaluation using MQM

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Ananya Mukherjee

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1 Introduction

This document defines the human evaluation guidelines for assessing machine-generated scientific figure annotations in a multilingual setting. The evaluation focuses on common figure types such as line plots, bar charts, and pie charts.

Unlike general image captioning, scientific figure description requires accurate interpretation of numerical values, axes, legends, and visual trends. To capture these requirements, we adopt an MQM-style framework that enables fine-grained error categorization and severity labeling.

These guidelines aim to ensure consistent, reliable, and reproducible human evaluation across languages, supporting the creation of a high-quality benchmark for figure-to-text generation research.

1.1 Task Description

Annotators are required to evaluate machine-generated figure annotations by performing two main tasks:

- **Pin-point the errors:** Identify and label all errors present in the generated annotation based on the predefined MQM-style error categories.
- **Assign Error Severity:** For each identified error, assign an appropriate severity level (Major or Minor) based on its impact on factual correctness, scientific interpretation, and overall usability of the annotation.

1.2 Role of a Human Evaluator

Human evaluators are responsible for assessing the quality and correctness of machine-generated scientific figure annotations. Their primary role is to carefully compare the generated description with the original figure and identify any mismatches, inaccuracies, or omissions.

Evaluators must apply the provided error taxonomy and severity definitions consistently, ensuring objective and evidence-based judgments. They should focus on factual accuracy, numerical correctness, visual grounding, and scientific appropriateness, rather than subjective writing style preferences.

By following these guidelines, evaluators help ensure reliable annotation quality, consistent evaluation across languages, and the creation of a trustworthy benchmark for figure-to-text generation research.

2 MQM Error Categories

This evaluation framework adopts a customized MQM-style error taxonomy designed for scientific figure annotation. Annotators are instructed to assign one or more error labels whenever applicable. [To Do: Add plots and corresponding examples from err analysis document]

2.1 Accuracy Errors

Accuracy errors refer to cases where the generated annotation does not correctly reflect the information presented in the figure.

Incorrect Numerical Value The annotation reports incorrect numerical values, percentages, or quantities.

Example: Reporting “60%” when the plot shows approximately “45%”.

Incorrect Trend Interpretation The annotation misrepresents the overall pattern or direction of change in the data.

Example: Claiming an upward trend when the values decrease after the second data point.

Incorrect Axis or Legend Interpretation The annotation incorrectly describes axis labels, units, or legend information.

Example: Stating that the x-axis represents temperature when it represents time.

Incorrect Label Mapping The annotation assigns incorrect category names or values to visual elements such as bars, lines, or pie segments.

Example: Assigning the blue bar to Method A when it corresponds to Method B.

2.2 Completeness Errors

Completeness errors occur when important information is missing or when content not present in the figure is added.

Missing Key Information The annotation omits essential elements such as major data points, axis labels, units, or key categories.

Example: Describing trends without mentioning the measurement units shown in the figure.

Hallucinated Content The annotation introduces information or visual elements that do not exist in the figure.

Example: Stating that the chart contains five categories when only four are present.

2.3 Clarity and Readability Errors

These errors affect the understandability and usability of the annotation.

Ambiguous Description The annotation contains vague or unclear references.

Example: Using phrases such as “this increases significantly” without specifying the subject.

Missing Takeaway The annotation fails to provide a meaningful summary of the main pattern or insight conveyed by the figure.

Example: Listing values without describing the overall trend or comparison.

Over-Generalization The annotation oversimplifies or exaggerates the visual information.

Example: Claiming that all methods perform similarly despite visible variation.

Overly Verbose Description The annotation contains unnecessary repetition or excessive detail that does not add informational value.

Poor Sentence Structure The annotation contains grammatical errors or awkward phrasing that reduces readability.

3 Severity Levels

Table 1 summarizes the severity levels used in the human evaluation framework. Each identified error is assigned either a Major or Minor label based on its impact on factual correctness, scientific reliability, and overall usability of the generated figure description. This severity distinction enables finer-grained analysis by separating critical content-related errors from stylistic and readability issues.

Table 1: Severity level definitions used in human evaluation.

Severity Level	Definition
Major	Errors that make the description factually unusable or misleading, or significantly degrade usability or scientific correctness.
Minor	Stylistic or readability issues that do not affect the core factual meaning of the annotation.

4 MQM Score Computation

The final MQM score for each generated annotation is computed by aggregating severity-weighted error penalties across all annotated error categories. For a given annotation, the total penalty score is calculated as:

$$\text{MQM} = \max \left(0, 100 - \sum_{i=1}^N w_{c_i, s_i} \right) \quad (1)$$

where:

- N is the total number of identified errors in the annotation,
- w_{c_i, s_i} is the weight associated with the i -th error, determined by its category c_i (Accuracy, Completeness, Clarity and Readability) and severity level s_i (Major or Minor), as defined in Table 2,
- The $\max(0, \cdot)$ ensures the score is clipped at 0 in cases where penalties exceed 100.

Lower MQM scores indicate higher annotation quality, with a score of zero representing an error-free description.

Table 2: Severity-weighted MQM error matrix used for human evaluation scoring.

Error Category	Major Weight	Minor Weight
Accuracy	5.0	2.0
Completeness	3.5	1.5
Clarity and Readability	2.0	1.0